

Assignment - 4 CSCU

Section - A

Q.1

Cyberbullying is the use of digital platforms to harass, threaten, or humiliate someone. Privacy settings help prevent it by restricting who can view posts, send messages or interact with a user's profile.

Q.2

- Unknown profiles with attractive offers.
- Urgent requests for money or personal data.
- Suspicious links
- Poor grammar or fake identities.

Q.3

- Unexpected attachments
- Strange file extensions (.exe, .zip)
- Unknown sender
- Urgent or threatening message tone.

Q.4

Email spoofing is when attackers fake the sender's address to appear trustworthy. Its purpose is to trick users into revealing sensitive data or downloading malware.

Q.5

Wi-Fi security protects wireless networks from unauthorized access.

Two methods :

- Use strong passwords (WPA2/WPA3 encryption).
- Hide SSID or enable firewall.

Section - B

- Q.1 Risks :
- Identity theft
 - Phishing attacks
 - Data misuse

How attackers create fake profiles :

- Use stolen photos
- Copy real profiles
- Add mutual friends

Preventive Measures :

- Verify profile authenticity
- Avoid accepting unknown requests
- Enable privacy controls
- Report suspicious accounts.

Q.2 Email based attacks

- Phishing : Fake emails to steal information
- Spoofing : Fake sender identity
- Malicious Attachments : Files containing malware.

Identification :

- Check sender address
- Avoid suspicious links
- Look for grammar errors.

Prevention :

- Use spam filters
- Avoid downloading unknown files
- Enable antivirus software.

Q.3

Common Attacks : • Unauthorized access
• Data interception
• Malware injection

Best Practices : • Use strong passwords
• Enable WPA 3 encryption
• Update router firmware.
• Disable remote access.
• Use MAC filtering

Section - C

Q.1

Technical Controls : • Spam filters and firewalls.
• Email encryption
• Multi-factor authentication (MFA)

User Awareness : • Training on phishing detection
• Safe email practices.

Monitoring Mechanisms : • Email logs analysis.
• Intrusion detection systems.
• Regular security audits.

Q.2

Online Backup : • Stored on cloud
• Accessible anywhere
• Risk : Hacking

Offline Backup : • Stored on external devices.
• Safer from cyberattacks.

- Limited accessibility.

Causes of Data Loss :

- Malware
- Hardware failure
- Human error

Complete Backup Plan :

- Regular automated backups
- Use both online and offline methods
- Encrypt backup data
- Test backups periodically

Q.3

How malware spreads :

- Untrusted downloads
- Email attachments
- Infected websites

Prevention Techniques :

- Use antivirus software.
- ~~Do not~~ Avoid unknown downloads
- ~~Update software~~ Keep system updated.

Endpoint Security Strategy :

- Install antivirus & firewall
- Enable automatic updates
- Use strong passwords
- Restrict admin access
- Monitor system activity